

Key

Note the quiz is front & back & any equations are given on a 2nd sheetPoints: ____/16_{16.5}

1. List 4 of the memorized polyatomic formulas with the correct charge for each. You don't need to write their names. (0.5 pt)

Slices

2. How many grams of H are in 20 g CH₄? (1 pt)

$$\frac{20 \text{ g CH}_4}{1} \cdot \frac{1 \text{ mol CH}_4}{16 \text{ g CH}_4} \cdot \frac{4 \text{ mol H}}{1 \text{ mol CH}_4} \cdot \frac{1 \text{ g}}{1 \text{ mol H}} = 5 \text{ g H}$$

- a. You need the molar mass (MM) of the above compound to solve the above question. Identify it here or above. (0.5 pt)

$$1 \times 12 + 4 \times 1 = 16 \text{ g/mol}$$

3. Determine the empirical formula of a compound that is 43.5 % phosphorus and 56.5% oxygen by mass. (1 pt)

$$\begin{array}{rcl} 43.5 \text{ g P} & 56.5 \text{ g O} & \\ \div 31 \text{ g/mol} & \div 16 \text{ g/mol} & \\ \hline 1.4 & 3.5 & \\ \times 100 & \times 100 & \\ \hline 14 & 35 & \end{array} \rightarrow \text{P}_{14}\text{O}_{35} \rightarrow \text{P}_2\text{O}_5$$

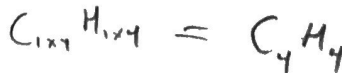
- a. When working with percents, it is easiest to assume you have ____ g as a total mass. identify the number that should go here. Getting the main part earns you these pt. Hint: it is also the normal boiling point of water. (0.5 pt)
- b. Which of the following is (are) not correct as an empirical formula? Circle all that will not be a proper empirical formula. (0.5 pt)

P_{0.5}O_{1.5}P₂O₂PO₂

4. A compound has an empirical formula of CH, and its molar mass is known to be 52 g/mol. What is its molecular formula? (1.5 pt)

$$\hookrightarrow \text{has a MM} = 13$$

$$52 \div 13 = 4$$



5. A DIPA is a mixture that has the same characteristics throughout and is approximately 7% ethanol by mass with the remainder water. Match the underlined terms to identify what the solution, solvent, and solute are. (1.5 pt)

$\begin{array}{ccc} \text{solution} & & \text{solute} \\ \downarrow & & \downarrow \\ \text{DIPA} & & \text{H}_2\text{O} \quad \text{Ethanol} \end{array}$

- a. What does the term aqueous signify for a solution? Briefly define it. (0.5 pt)

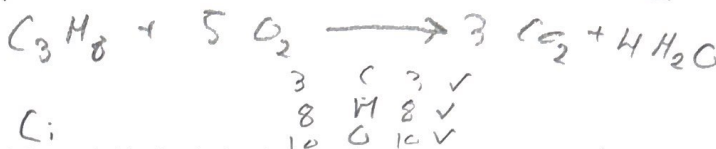
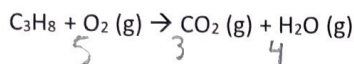
$\hookrightarrow$ H₂O is the solvent / is present.
Things/solutes dissolved in H₂O.

→ 0.355 L

6. A 355-mL soft drink sample contains 0.133 mol of sucrose (table sugar). What is the molar concentration (molarity) of sucrose in the beverage? (1 pt)

$$\frac{0.133 \text{ mol}}{0.355 \text{ L}} = 0.375 \text{ M}$$

- a. Identify the two units that make up molarity. Getting the main part right earns you this. (0.5 pt)
7. Balance the following chemical equation. (1 pt)



8. What is the resulting volume when 30 mL of a 1.35 M NaI (aq) solution is diluted with pure water to make a 0.07 M NaI (aq) solution? (1 pt)

$$C_i V_i = C_f V_f \rightarrow V_f = \frac{C_i V_i}{C_f} = \frac{1.35 \text{ M} \cdot 30 \text{ mL}}{0.07 \text{ M}} = 579 \text{ mL}$$

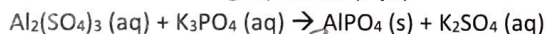
- a. Identify which concentration above corresponds to the more concentrated solution. (0.5 pt)

1.35

- b. What amount of pure water was necessary for this dilution? (0.5 pt)

$$579 - 30 = 549 \text{ mL}$$

9. Balance the following equation. (1 pt)

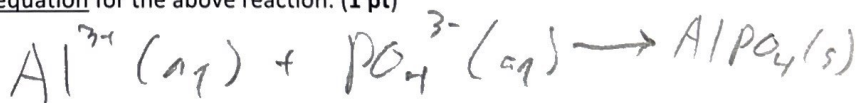


- a. Identify which compound above is (are) the precipitate(s). all identifications are graded. (0.5 pt)

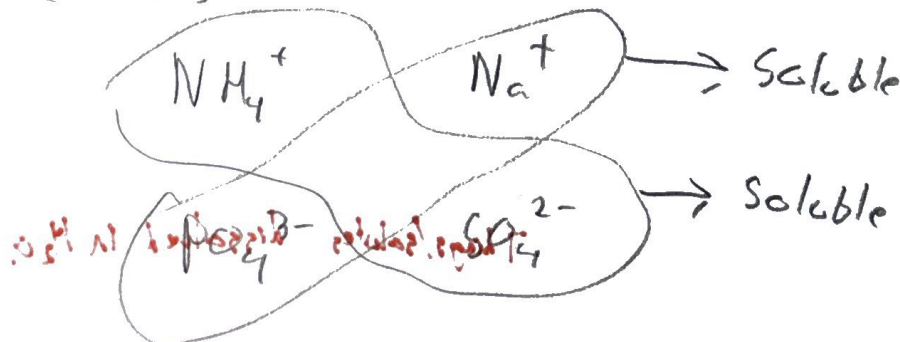
AlPO_4

- b. Write the balanced net ionic equation for the above reaction. (1 pt)

if coeff. of 2
(-3)



10. Complete the balanced molecular chemical equation for the precipitation reaction below. If no reaction occurs, write NR after the reaction arrow. Be sure to include all proper phases for all species in the reaction. You should show some work using the solubility tables on the equation sheet to explain your answer. (2 pt)



• if → (aq) + (aq)
⊕
• if NR w/ no work
(-1.5)

SHOW YOUR WORK

Quiz 6 Nov 6th 2025 Chem 1127

Name (Full First then Last or -1 pts):

Key

Note the quiz is front & back & any equations are given at the end.

Points: ____/16₁₇

1. What is the colloquial (common or everyday) name of NaN_3 and why is it called that? (1 pt)

Smile - named due to its use to kill pests

2. What mass of water is produced when 400 g C_8H_{18} is combusted via the below equation? (1 pt)



$$\text{MM} = 2 \times 1 + 16 = 18 \text{g/mol}$$

$$\text{MM} = 8 \times 12 + 18 = 114 \text{g/mol}$$

$$\frac{400 \text{g C}_8\text{H}_{18}}{1} \times \frac{1 \text{mol C}_8\text{H}_{18}}{114 \text{g C}_8\text{H}_{18}} \times \frac{18 \text{g/mol H}_2\text{O}}{2 \text{mol C}_8\text{H}_{18}} \times \frac{18 \text{g H}_2\text{O}}{1 \text{mol H}_2\text{O}} = 568 \text{g H}_2\text{O}$$

- a. What word is implied to be between each coefficient & chemical formula above? Getting the main part write earns you this pt. (0.5 pt)

moles (or molecules)

- b. The above problem requires you to determine molar masses (MM). Clearly identify one of the compounds and its correct MM. (0.5 pt)

✓

- c. What phase of matter (s, l, g, or aq) is CO_2 commonly observed as? (0.5 pt)

gas

3. A species that is an acid Donates protons, and will produce Hydronium upon reaction with water. while a species that is reduced Accepts/gains electrons. Correctly fill in the blanks with the below words. A word may be used multiple times. Note if you would rather use a different word, you may, but the words below will work. (2 pt)

Hydroxide(s)

Proton(s)

Electron(s)

Neutron(s)

Hydronium(s)

Donates

Accepts

Transmutes

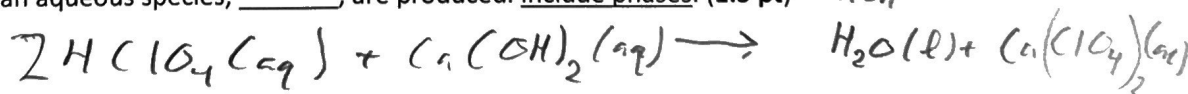
Capitalizes

"GER" - gaining e^- is reduction.

4. List 4 strong bases in the space below. Names or formula are fine. (1 pt)

Slit,

5. Write & balance the neutralization reaction/equation between HClO_4 (aq) & $\text{Ca}(\text{OH})_2$ (aq). Note, water and an aqueous species, _____, are produced. Include phases. (1.5 pt)



6. Identify the oxidation number (state) for each element below. Note, if the same element is given twice, it may have different oxidation numbers in each. (2.5 pt)

V_2O_5

O_2

Cl^-

+5

0

-1

H-Li - metals generally positive
-1 +1

- a. Oxidation numbers are like _____. Circle the phrase below that goes in the blank.

(0.5 pt)

Electronegativity

Ionization Energy

Ionic Charge

Energy Transfer

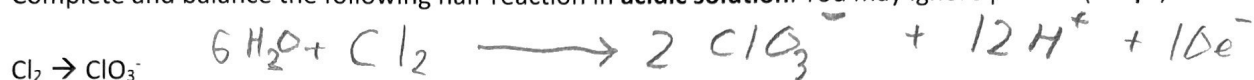
SHOW YOUR WORK



7. Is $2 \text{H}_2\text{O}_2(\text{aq}) \rightarrow 2 \text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ a redox, precipitation, or acid-base reaction? Circle the correct one and briefly explain your answer. (1.5 pt)

$\rightarrow e^-$ are transferred changing oxidation state of oxygen

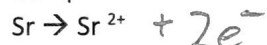
8. Complete and balance the following half-reaction in **acidic solution**. You may ignore phases. (1.5 pt)



e^- are lost

- a. Identify whether this is a reduction or oxidation half-reaction. You must have at least attempted the main part to be able to earn this point. (0.5 pt)

9. Complete and balance the following half-reaction. You may ignore phases. (1 pt)



10. Briefly describe an issue with a chemical reaction that you want completed, if it has a 2% yield. (1 pt)

Means side reactions are occurring, & the desired reaction is inefficient!

11. Atoms that have the same number of protons, but different numbers of neutrons are called _____. inefficiency!
Circle the word here that completes the phrase. (0.5 pt)

Ions

Compounds

Isomers

Molecules

Isotopes